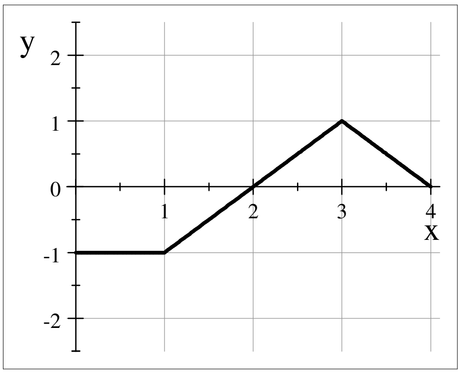
WTW 158 Worksheet 10 on Integration

A. TEXTBOOK PROBLEMS
Integral and Area problems:
p 375 no 4
p 388 no 33, 37, 39, 51
p 399 no 45, 53, 55
Integration problems:
p 399 no 19, 25, 29, 39, 43,
p 408 no 9 - 17 (odd)
p 418no 29 - 45 (odd), 63, 65, 69)

B. MORE PROBLEMS
1. 55 Integration Challenge, available on ClickUP under Worksheets
2. Technique Mastering Problems p 49 N - Integration by inspection

C. TEST QUESTIONS
1.



For the function f on the interval $[0,4]$, of which the graph is shown:
1.1 $\int_0^4 f(x)\, dx =$

(a)	-0.5	(b)	-1.5	(c)	2.5	(d)	0.5
(e)	3	(f)	None of these				

1.2. The area enclosed between the curve and the x-axis on $[0,4]$ is given by

(a)	-0.5	(b)	-1.5	(c)	2.5	(d)	0.5
(e)	3	(f)	None of these				

2. Find the area bounded by the curve $y = 2x - x^2$, the x-axis and the lines $x = -1$ and $x = 2$.

D. SELECTED ANSWERS
A. Integral and area problems p 375 no 4 $RHS = \frac{\pi}{8}(\sin\frac{\pi}{8} + \frac{1}{\sqrt{2}} + \sin\frac{3\pi}{8} + 1)$,
$LHS = \frac{\pi}{8}(0 + \sin\frac{\pi}{8} + \frac{1}{\sqrt{2}} + \sin\frac{3\pi}{8})$
p 388 See p A93; p 399 See p A93
Integration problems p399 See p A93; p 408 See p A94; p 418 See p.A94
B. 1. See ClickUP 2. See ClickUP
C. 1.1 [a] 1.2. 2.5 2. $\frac{8}{3}$
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